Supplementary Material for "Survival-Time Risk Beyond Low-Order Channel Models: Identifiability and Scheduling"

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S1. Overview

This supplement collects material that supports the main paper but does not fit its six pages. Results of the main paper are cited by their numbers there, e.g., Lemma 1, Theorem 1 or Fig. 4; numbering with the prefix "S" refers to this document. Notation follows the main paper. Section S2 illustrates the system model, gives the design of π_L in full and represents the risk of any policy through sampled loss patterns. Section S3 gives complete proofs and additional theory. Sections S4 and S5 report further numerical results behind the claims of the main paper, and Section S6 maps each claim to the script that produces it. All result tables and data figures in this supplement are generated from the stored results by `code/make_figs.py` and `code/make_supplement.py`.

S2. System Model and Low-Order Design

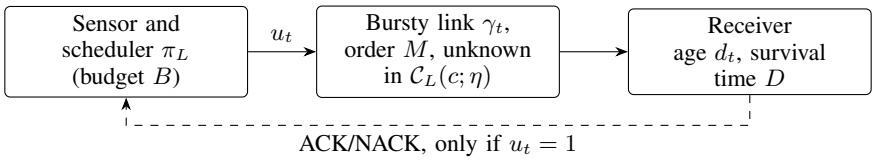


Fig. S1. System model. The latent loss process evolves in every slot; the scheduler observes it only in slots where it transmits.

Fig. S1 shows the setting of Section II. The scheduler π_L minimizes the risk computed under the fitted L-th order model. For $L = 1$ let $q_o = c(o1)/c(o)$ and $f(x) = q_0 + (q_1 - q_0)x$. If the last observed outcome o was observed δ slots ago, the fitted model predicts success with probability $f^{\delta-1}(q_o)$; before any observation it predicts $c(1)$, and the counter δ is not advanced. With $V_{T+1} \equiv 0$, $F(d', z) = 1$ if $d' \geq D$ and $F(d', z) = z$ otherwise, and $q = f^{\delta-1}(q_o)$, the design solves

$$\begin{aligned} V_t(d, r, o, \delta) &= \min\{V^0, V^1\}, \qquad V^0 = F\big(d{+}1, V_{t+1}(d{+}1, r, o, \delta{+}1)\big), \\ V^1 &= q\, V_{t+1}(0, r{-}1, 1, 1) + (1{-}q)\, F\big(d{+}1, V_{t+1}(d{+}1, r{-}1, 0, 1)\big), \end{aligned} \tag{S1}$$

where V^1 requires $r \geq 1$ and ties are broken toward idling. The nominal risk R_{nom} is $V_1(0, B, \varnothing, 0)$. If $B \geq T$, attempting in every slot minimizes the age pathwise on every channel, since an idle slot can only delay a reset. Where a failure is still possible, idling leads, with positive probability under the fit, to a failure that an attempt would avoid, because the fit is η-positive whenever $\mathcal{C}_L(c;\eta)$ is nonempty. Hence π_L attempts in exactly these slots, and its risk equals that of dense attempts on every channel. The same recursion with an arbitrary recovery curve $\Pr(\gamma_{t+\delta} = 1 \mid o)$ in place of $f^{\delta-1}(q_o)$ defines the redesign π_{rec} of Section S5.2.

Proposition S1 (Sampled-pattern representation)**.** *For every deterministic policy π with budget B under ACK/NACK feedback, there is a finite set $\mathcal{H}_\pi$ of attempt-slot tuples $\mathbf{s} = (s_1 < \cdots < s_j)$ and outcome tuples $\mathbf{o} \in \{0,1\}^j$, determined by π alone, such that for every channel $\mathcal{M}$*

$$R_{\mathcal{M}}(\pi) = \sum_{(\mathbf{s},\mathbf{o}) \in \mathcal{H}_\pi} \Pr_{\mathcal{M}}(\gamma_{s_1} = o_1, \ldots, \gamma_{s_j} = o_j). \tag{S2}$$

Proof: The controller's information, the age and the remaining budget are functions of the attempted outcomes only. Enumerating the ACK/NACK histories up to the failure slot partitions $\{\tau \leq T\}$ into disjoint cylinder events on attempted slots. ∎

The risk is therefore a linear functional of gapped pattern probabilities, and the task selects which patterns enter. It is identifiable from c if all of them are. This sufficient condition fails in general: with dense attempts and $D > L + 1$, for instance, every failure pattern fixes at least D consecutive outcomes. The closed-loop bound Eq. (15) makes the selected patterns explicit.

S3. Proofs and Additional Theory

S3.1 Proof of Lemma 1

A stationary M-th order chain with positive context law is equivalent to a shift-consistent $(M{+}1)$-block law ν, i.e., one whose left and right M-marginals coincide (both equal the context law μ): given such ν, the transition probabilities $p(w) = \nu(w1)/\mu(w)$ started from μ define a stationary chain with block law ν, and conversely. For $M = L{+}1$ the $(L{+}1)$-block law of the chain is μ itself, so compatibility means that both M-marginals of ν equal c. For a middle word $m \in \{0,1\}^L$ the four entries $\nu(a\,m\,a')$, $a, a' \in \{0,1\}$, form a 2×2 table with row sums $c(am)$ and column sums $c(ma')$; distinct middle words involve disjoint entries. Each table has one free entry $x_m = \nu(0m0)$, and the others follow as stated in the lemma. Nonnegativity yields the Fréchet interval $\max\{0, c(m0) - c(1m)\} \le x_m \le \min\{c(0m), c(m0)\}$. The constraint $\eta \le p(am) \le 1 - \eta$ reads $x_m \in [\eta c(0m), (1-\eta)c(0m)]$ for $a = 0$ and $c(1m) - c(m0) + x_m \in [\eta c(1m), (1-\eta)c(1m)]$ for $a = 1$; the intersection is $[x_m^{\mathrm{lo}}, x_m^{\mathrm{hi}}]$ and implies nonnegativity. Since the tables are independent, the class is the product of these intervals, and the map is affine.

The entropy rate of the chain is $\bar{H} = H(\gamma_{M+1} \mid \gamma_{1:M}) = H(\gamma_{L+2} \mid \gamma_1, \gamma_{2:L+1}) \le H(\gamma_{L+2} \mid \gamma_{2:L+1})$, and the right side depends only on c. Equality holds iff γ_1 and γ_{L+2} are conditionally independent given every middle word, i.e., $\nu(a\,m\,a') = c(am)c(ma')/c(m)$, which gives x_m^{nom}; this is the fitted L-th order model, whose transition $\Pr(a' \mid m) = c(ma')/c(m)$ ignores a. Uniqueness follows from strict concavity of the conditional entropy in each table. In the running example, $x_0 = \nu(000) \in [\frac{11}{250}, \frac{57}{500}]$ and $x_1 = \nu(010) \in [\frac{1}{250}, \frac{11}{250}]$, and the class is parameterized by (ρ, b) through $p(00) = 1 - \rho$, $p(01) = b$, $p(10) = 1 - \frac{3}{2}(1-\rho)$, and $p(11) = 1 - \frac{b}{9}$.

S3.2 Proof of Theorem 1 and Eq. (12)

Continue the age process past D (the schedule is open-loop, so attempts do not depend on failure) and let $I_t = \mathbf{1}\{d_t = D\}$. The age increases by one per slot until it is reset, so it cannot jump over D, and $\{\tau \le T\} = \{\sum_{t \le T} I_t \ge 1\}$. Moreover, $I_t = 1$ iff the last reset before t occurred at $t - D \in \mathcal{S}_0$ and no attempt in $(t-D, t]$ succeeded, i.e., iff A_{t-D} occurs; hence $\mathbb{E}\sum_t I_t = \Lambda_{\mathcal{S}}$, and the union bound gives $R \le \Lambda_{\mathcal{S}}$. This step uses neither the Markov property nor stationarity.

For two hits $t < t'$, the reset at $t' - D$ comes after t: if $t' - D \le t$, the age at t would be $t - (t' - D) < D$ or 0. Hence $t' - D > t$, so I_t is measurable with respect to $\mathcal{F}_{t'-D-1} = \sigma(S_1, \gamma_{1:t'-D-1})$, and $t' - D \in \mathcal{S}$ because $t' - D > t \ge D \ge 1$. By the Markov property, $\mathbb{E}[I_{t'} \mid \mathcal{F}_{t'-D-1}] = \Pr(A_{t'-D} \mid S_{t'-D}) \le \kappa_{\mathcal{S}}$. For fixed $t = s + D$ there are at most $J_{\mathcal{S}}$ admissible t', so $\sum_{t<t'} \mathbb{E}[I_t I_{t'}] \le J_{\mathcal{S}} \kappa_{\mathcal{S}} \Lambda_{\mathcal{S}}$, and the second Bonferroni inequality $\Pr(\cup E_i) \ge \sum \Pr(E_i) - \sum_{i<j} \Pr(E_i E_j)$ yields the lower bound.

For dense attempts, $A_0 = \{0^D\}$ and $A_s = \{\gamma_s = 1\} \cap \{0^D \text{ on } (s, s+D]\}$ for $s = 1, \dots, n$. Since $\Pr(0^{M+1}) = x_{0^L} = \rho\, c(0^M)$ and every further loss after M losses has probability ρ, $\Pr(0^D) = c(0^M)\rho^k$ and, by stationarity, $\Pr(10^D) = \Pr(0^D) - \Pr(0^{D+1}) = c(0^M)(1-\rho)\rho^k$. Given $S_s = w$, A_s requires a success followed by D losses, so $\Pr(A_s \mid w) \le \rho^k \max_{v: v_M = 1} \Pr(0^M \mid v)$, and $J = (T - 2D)^+ \le n$; this gives $J\kappa \le \theta_{\mathcal{M}} \Lambda$. In the running example ($M = 2$), the contexts ending in a success are 01 and 11, and $\Pr(00 \mid v) = (1 - p(v))(1 - p(10))$ with $1 - p(10) = \frac{3}{2}(1-\rho)$, $1 - p(01) = 1 - b$ and $1 - p(11) = \frac{b}{9}$; dividing by $c(00)(1-\rho) = 0.12(1-\rho)$ gives $\theta_{\mathcal{M}} = 12.5 \max\{1 - b, \frac{b}{9}\} \le 6.9$.

S3.3 Details of Theorem 2

With $\varphi(\rho) = \rho^k[1 + n(1-\rho)]$ as in Theorem 2, $\varphi'(\rho) = \rho^{k-1}[k(n+1) - (k+1)n\rho]$, so φ is increasing for $n \le k$ and unimodal with peak $\rho^\star = k(n+1)/[(k+1)n]$ otherwise; $\rho^\star \in (k/(k+1), 1)$ and decreases in n. The fitted model is the worst compatible channel only if $\rho_{\mathrm{nom}} = \rho_w$; in the running example this happens, e.g., for $k = 1$, $n = 5$ ($D = 3$, $T = 8$), where $\rho^\star = \frac{6}{10} = \rho_{\mathrm{nom}}$. Generically the fitted model is not the worst case.

Required precision. $\partial \log \Lambda / \partial \rho = k/\rho - n/[1 + n(1-\rho)]$, which is 14.3 at $\rho_{\mathrm{nom}} = 0.6$ for $T = 60$, $D = 12$; an error of $\Delta\rho$ multiplies the risk by about $e^{14.3\,\Delta\rho}$, so a factor-two accuracy requires $|\Delta\rho| \lesssim 0.049$, and the required precision grows linearly in D. In the running example, $\rho = \Pr(0 \mid 00)$ is estimated from the $\approx c(00)N = 0.12N$ occurrences of the context 00 in N slots; ignoring the dependence between occurrences, a two-standard-error accuracy of 0.049 needs $N \approx 3.4 \times 10^3$ slots. The estimation problem becomes hard only when the order is unknown.

S3.4 Proposition 1 for arbitrary D

The proof of Proposition 1 does not need $g \mid D$. For $s \in g\mathbb{N}_0$, the attempts in $(s, s+D]$ are $s+g, \dots, s+D_g g$ with $D_g = \lfloor D/g \rfloor$, and $s + D \le T$ holds for $n_g = \lfloor (T-D)/g \rfloor$ values $s \ge g$. Hence $\Lambda_{\mathcal{S}_g} = \Pr^{(g)}(0^{D_g})[1 + n_g(1 - \rho^{(g)}_{D_g})]$ for every $D \ge g$; only the failure slot ($s + D$ instead of $s + D_g g$) differs from dense attempts on $\gamma^{(g)}$. For M-th order chains and primitive Q_g, $\rho_j^{(g)} \to \sigma_g$ as $j \to \infty$, because $\Pr^{(g)}(0^j) = \alpha^\top Q_g^{j-1} \mathbf{1}$ for a fixed nonnegative vector α.

S3.5 Decay rates of spaced attempts

Proposition S2 (Decay exponents)**.** *With Q_g and σ_g as in Section IV, $\sigma_1 = \rho$, $\sigma_g \geq \rho^g$, and Q_g is entrywise positive for $g \geq M$. If $\log(1 + T - D) = o(D)$, and $g = 1$ or Q_g is primitive, then Eq. (14) holds.*

Proof: Let $D_g = \lfloor D/g \rfloor$. For $s \in \mathcal{S}_g$, define $\alpha_s(w) = \Pr(\gamma_s = 1, \gamma_{s+g} = 0, S_{s+g+1} = w)$ on loss-ending contexts; then $\Pr(A_s) = \alpha_s^\top Q_g^{D_g - 1}\mathbf{1}$, and similarly for A_0. By stationarity α_s does not depend on s, and $\Lambda_{\mathcal{S}_g}$ has at most $n/g + 1$ such terms. If Q_g is primitive, Perron–Frobenius gives $Q_g^j = \sigma_g^j(\xi\zeta^\top + o(1))$ with positive ξ, ζ, hence $\Pr(A_s) = \Theta(\sigma_g^{D_g})$ and $\log \Lambda_{\mathcal{S}_g} = D_g \log \sigma_g + O(\log n) + O(1)$. Since every context has η-positive success probability, Q_g is strictly substochastic and $\sigma_g < 1$. The second-order term satisfies $J\kappa \leq (n/g) \max_w e_w^\top Q_g^{D_g - 1}\mathbf{1} = O(n\sigma_g^{D_g}) \to 0$ when $\log n = o(D)$, so $\log R = \log \Lambda + o(1)$ and $\frac{1}{D} \log R \to \frac{1}{g} \log \sigma_g$. For $g = 1$, Q_1 is not primitive, but Eq. (12) gives Λ in closed form and $\frac{1}{D} \log \Lambda \to \log \rho$. In the loss graph of Q_1 the only cycle is the self-loop at 0^M, so $\sigma_1 = \rho$. Finally, $\sigma_g \geq Q_g(0^M, 0^M) \geq \rho^g$, since the spectral radius of a nonnegative matrix dominates every diagonal entry, and for $g \geq M$ all $g-1$ intermediate outcomes are free, so every loss-ending context is reached with η-positive probability. ∎

For $M = 2$, $g = 2$ and contexts ordered $(00, 10)$,

$$Q_2 = \begin{pmatrix} \rho^2 & (1-\rho)(1-b) \\ (1 - p(10))\rho & p(10)(1-b) \end{pmatrix}, \tag{S3}$$

whose second column (loss–success–loss) holds the relapse terms. For C_1 and C_2, $\sigma_2 = 0.239$ and 0.433; exact risks at $n = 48$ give log-ratio slopes 0.1823 and 0.2986 per unit of D, against $\log(\rho(C_1)/\rho(C_2)) = 0.1823$ and $\frac{1}{2}\log(\sigma_2(C_2)/\sigma_2(C_1)) = 0.2986$, and the reversal holds for every even D from 4 to 40. For large D this is a criterion: if $\rho(C_1) > \rho(C_2)$ but $\sigma_g(C_1) < \sigma_g(C_2)$, with Q_g primitive for both and $\log n = o(D)$, dense and g-spaced attempts rank the two channels oppositely, with log-ratios growing at these slopes.

S3.6 Closed-loop bound: proof and the missing lower bound

Proposition S3 (Closed-loop bound)**.** *For every deterministic policy π under ACK/NACK feedback, extended arbitrarily beyond failure, $R(\pi) \leq \Lambda_\pi$ with Λ_π as in Eq. (15).*

Proof: Extend π beyond failure by any fixed rule. After a success at s the controller state is a function of the observation history h up to s; for π_L it is $(d, o, \delta) = (0, 1, 1)$ together with the remaining budget r. As long as the attempts fail, the observations are NACKs, so the decisions in $(s, s+D]$ are a deterministic function of (s, h): the retry pattern $\mathcal{R}_\pi(s, h)$. Hence $I_{s+D} = \mathbf{1}\{\text{success at } s\}\, \mathbf{1}\{\gamma_j = 0 \ \forall j \in \mathcal{R}_\pi(s, h)\}$. Since h is a function of $\gamma_{1:s}$, conditioning on (h, S_{s+1}) and using the Markov property gives Eq. (15); the union bound gives $R \leq \Lambda_\pi$. For the start ($s = 0$), h is empty, $r = B$ and $S_1 \sim \mu$. ∎

The second Bonferroni inequality still yields $R \geq \Lambda_\pi(1 - \omega_\pi)$, where ω_π is the largest expected number of further hits, given the state at a hit. For open-loop schedules $\omega_\pi \leq J\kappa$, which recovers Theorem 1. For budgeted closed-loop policies, however, ω_π is of order one: a success that spends the last attempt makes the next hit certain, and the continuation after a failure can create further resets. Table S1 evaluates both bounds for π_L ($T = 60$, $B = 15$, $D = 12$ unless stated). It reports $\omega_\pi \approx 1.7$, so this bound is vacuous, while the upper bound is tight: over a 9×5 grid of the class, $\Lambda_\pi/R \leq 1.30$. The exact second factorial moment $\sum_{t<t'} \mathbb{E}[I_t I_{t'}]$ gives a valid lower bound but is not simpler to compute than R itself.

TABLE S1
CLOSED-LOOP BOUND Λ_π FOR π_L

channel	$R(\pi_L)$	Λ_π	Λ_π/R	ω_π	decomposition error
nominal	5.97×10^{-2}	7.41×10^{-2}	1.242	1.66	9×10^{-16}
C_1	1.30×10^{-2}	1.51×10^{-2}	1.161	1.72	1×10^{-16}
C_2	6.04×10^{-2}	7.00×10^{-2}	1.159	1.72	4×10^{-16}
C_2, $D = 16$	1.51×10^{-2}	1.62×10^{-2}	1.069	0.95	2×10^{-16}
C_2, $D = 20$	4.16×10^{-3}	4.26×10^{-3}	1.024	0.63	0

S3.7 No task-agnostic burstiness index

Corollary S1. *No function ψ on $\mathcal{C}_L(c; \eta)$, such as the burst-length tail, a lag-g correlation with $g > L$, ρ or a Gilbert–Elliott parameter, satisfies $R_{\mathcal{M}}(\pi_L) < R_{\mathcal{M}'}(\pi_L) \Rightarrow \psi(\mathcal{M}) < \psi(\mathcal{M}')$ for every task (T, D, B).*

Proof: Item 1) of Proposition 2 forces $\psi(C_2) < \psi(C_1)$, and item 3) forces $\psi(C_1) < \psi(C_2)$. ∎

S3.8 Order mismatch $M \geq L + 2$

Remark S1. *For $M \geq L + 2$ the compatible class is a general polytope, and $y = \Pr(0^M)$ is no longer fixed by c. With $x = \Pr(0^{M+1})$, Eq. (12) generalizes to $\Lambda = y\,\varphi(x/y)$, which is homogeneous of degree one in (y, x). For fixed $\rho = x/y$, Λ is proportional to y, so the maximum over the class lies where a ray of slope ρ leaves the two-dimensional projection of the polytope onto (y, x), i.e., on its boundary; the projection follows from linear programs. For the running example with $M = 3$ instead of $M = 2$ ($T = 60$, $D = 12$), $y \in [0.044, 0.114]$ (nominal 0.072), and a grid search over the projected polygon gives $\Lambda_{\max} = 0.254$ (0.253 for $M = 2$) at $y = 0.112$ and $x/y = 0.928$, where the exact dense risk is 0.240. Since the $M = 3$ class contains the $M = 2$ class, whose dense worst case is 0.242, the $M = 3$ dense worst case lies in $[0.242, 0.254]$. The closed-loop worst case over the polytope remains open.*

S4. Additional Results for the Running Example

S4.1 Dense attempts

Table S2 compares the worst-case persistence of Theorem 2 with exact maximizers over a grid of 1201 values of ρ at $b = 0.7$. The maximizer of Λ matches ρ_w to grid resolution, and the maximizer of R deviates slightly where R saturates; over all cases, the two-sided bound is violated by at most 1.4×10^{-16}. Fig. S2 shows the phase diagram: for long missions the worst channel is interior, and once $D \geq D^\star = 21$ the cap $\rho_{\mathrm{hi}} = 0.95$ binds for every mission length. The right panel shows that the risk ratios R^+/R_{nom} and R^+/R^- grow geometrically in D, as predicted by Λ, until R^+ saturates. Table S3 repeats the comparison for a third-order class ($L = 2$, $M = 3$), a four-dimensional box with $\rho \in [0.488, 0.950]$ and $\rho_{\mathrm{nom}} = 0.650$. Its column "spread" is the largest relative change of the dense risk when the three relapse coordinates are redrawn at fixed ρ; it is at most 2.7%. Over random channels and dense, periodic and irregular open-loop schedules ($T = 60$, $D = 12$), the two-sided bound of Theorem 1 was never violated, and its gap $(\Lambda_{\mathcal{S}} - R)/R$ never exceeded 16%.

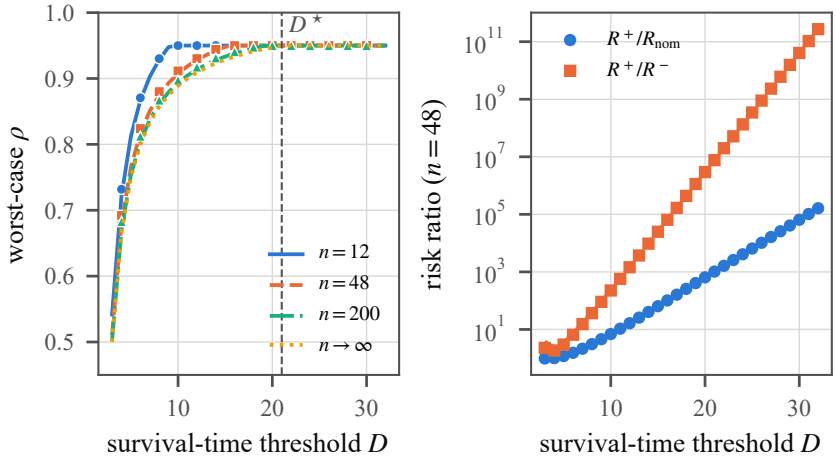


Fig. S2. Left: worst-case persistence ρ_w (lines: Theorem 2; markers: exact risk maxima) versus D for $n = T - D$; the cap $\rho_{\mathrm{hi}} = 0.95$ is active for all n beyond $D^\star = 21$. Right: exact risk ratios for $n = 48$ (markers) and their predictions from Λ (lines).

TABLE S2
DENSE ATTEMPTS OVER THE COMPATIBLE CLASS OF THE RUNNING EXAMPLE

T	D	ρ_w (Thm.)	argmax Λ	argmax R	R^+	R^-	R_{nom}	R^+/R_{nom}
6	6	0.950	0.950	0.950	9.8×10^{-2}	2.2×10^{-3}	1.6×10^{-2}	6.3
12	6	0.933	0.933	0.933	1.3×10^{-1}	1.0×10^{-2}	5.3×10^{-2}	2.4
30	6	0.833	0.833	0.840	2.7×10^{-1}	3.5×10^{-2}	1.6×10^{-1}	1.7
60	6	0.815	0.815	0.822	4.7×10^{-1}	7.4×10^{-2}	3.1×10^{-1}	1.5
120	6	0.807	0.807	0.815	7.2×10^{-1}	1.5×10^{-1}	5.3×10^{-1}	1.4
30	10	0.933	0.933	0.935	1.6×10^{-1}	5.4×10^{-4}	1.8×10^{-2}	8.8
60	10	0.907	0.907	0.910	2.9×10^{-1}	1.3×10^{-3}	4.2×10^{-2}	6.8
200	10	0.894	0.894	0.898	6.8×10^{-1}	4.7×10^{-3}	1.4×10^{-1}	4.7
100	15	0.939	0.939	0.942	3.0×10^{-1}	1.4×10^{-5}	5.5×10^{-3}	55
300	15	0.932	0.932	0.935	6.6×10^{-1}	4.7×10^{-5}	1.8×10^{-2}	37
100	24	0.950	0.950	0.950	1.8×10^{-1}	1.5×10^{-9}	5.0×10^{-5}	$3.6k$
400	24	0.950	0.950	0.950	5.6×10^{-1}	7.4×10^{-9}	2.4×10^{-4}	$2.4k$

TABLE S3
DENSE ATTEMPTS OVER THE THIRD-ORDER CLASS ($L = 2$, $M = 3$)

T	D	ρ_w (Thm.)	argmax R	R^+	R^-	R_{nom}	spread
8	8	0.950	0.950	4.6×10^{-2}	1.7×10^{-3}	7.0×10^{-3}	0.0%
60	8	0.849	0.847	2.2×10^{-1}	4.5×10^{-2}	1.3×10^{-1}	2.7%
200	8	0.838	0.842	5.7×10^{-1}	1.5×10^{-1}	3.9×10^{-1}	2.4%
100	12	0.910	0.913	2.1×10^{-1}	4.4×10^{-3}	3.9×10^{-2}	1.6%
400	12	0.902	0.899	6.2×10^{-1}	1.9×10^{-2}	1.6×10^{-1}	1.5%
150	30	0.950	0.950	1.0×10^{-1}	1.5×10^{-8}	2.3×10^{-5}	0.5%

S4.2 Budgeted attempts, certificates and the positivity bound

Table S4 is the closed-loop sweep over the budget summarized in Fig. 3 ($T = 60$, $D = 12$; a grid of 47×21 channels in (ρ, b) with resolution 0.013 in ρ). As B decreases, the worst channel moves toward less persistent channels with frequent relapses ($b = 0.45$), and the best channel, at ρ_{lo}, moves from $b = 0.45$ ($B = 60$) to $b = 0.95$ (all smaller budgets). At $\rho = \frac{11}{30}$, switching b from 0.95 to 0.45 leaves the dense risk nearly unchanged but raises the risk of periodic attempts every four slots by $3.88\times$, every two slots by $34.6\times$, and that of π_L with $B = 15$ by $9.05\times$; at $\rho = 0.95$ the same switch changes the dense, every-four-slots and π_L risks by at most 19%. Table S5 lists the certificates discussed in Section V ($T = 60$). The interval relaxation, i.e., the rectangular robust MDP, lets each context's success probability range over the projection of the box and be chosen adversarially in every slot; the time-invariant corner fixes the worst corner of these intervals for the whole mission. Table S6 varies the positivity bound η ($T = 60$, $D = 12$, $R_{\mathrm{nom}} = 0.0146$): the dense worst case is interior, so it barely moves.

TABLE S4
CLOSED-LOOP RISK OF π_L OVER THE COMPATIBLE CLASS

B	R_{nom}	R^+	at (ρ, b)	R^-	R^+/R_{nom}
60	1.5×10^{-2}	0.241	(0.925, 0.95)	1.7×10^{-4}	16.5
30	2.2×10^{-2}	0.254	(0.925, 0.45)	4.7×10^{-4}	11.4
20	3.7×10^{-2}	0.269	(0.912, 0.45)	1.7×10^{-3}	7.3
15	6.0×10^{-2}	0.286	(0.899, 0.45)	6.7×10^{-3}	4.8
12	9.3×10^{-2}	0.303	(0.887, 0.45)	2.2×10^{-2}	3.3

TABLE S5
RISK CERTIFICATES FOR THE RUNNING EXAMPLE

Task	R_{nom}	R^+ (grid)	box $\Lambda_{\max}$	interval relax.	time-inv. corner
dense, $D = 12$	1.5×10^{-2}	0.242	0.253	0.973	0.967
dense, $D = 20$	2.1×10^{-4}	0.142	0.143	0.795	0.782
$B = 15$, $D = 12$	6.0×10^{-2}	0.286	–	0.985	0.982

TABLE S6
SENSITIVITY OF THE DENSE WORST CASE TO η

η	$[\rho_{\mathrm{lo}}, \rho_{\mathrm{hi}}]$	worst ρ	R^+	R^-	R^+/R_{nom}
0.01	$[0.340, 0.990]$	0.931	0.239	8.1×10^{-5}	16.3
0.02	$[0.347, 0.980]$	0.931	0.239	9.7×10^{-5}	16.4
0.05	$[0.367, 0.950]$	0.931	0.240	1.7×10^{-4}	16.5
0.1	$[0.400, 0.900]$	0.900	0.230	3.7×10^{-4}	15.8

S4.3 Exact certificate of Proposition 2

The risks of Proposition 2 were computed in exact rational arithmetic (`fractions.Fraction`), including the design of π_L. The full fractions are:

- dense attempts, $R(C_1) \approx 9.096922\times10^{-4}$:
  `28764046942402247011864795479079374618670119590699773794325455937275452419409017556257663661031306070 09`

630880007278304587839377494 3/3161953707645960904043087978513551661308156326413154602050781250000000000

under ACK/NACK feedback, which requires a dynamic program over beliefs, is not computed here; its risk lies between that of the genie and that of the better of π_L and π_{rec}. On the measured links of Section S5.5, the redesign with the empirical recovery curve improves on π_L by a median factor of 1.015 and at most 1.27.

TABLE S9
DESIGNING WITH THE TRUE RECOVERY CURVE

channel	ρ	b	$R(\pi_L)$	$R(\pi_{\mathrm{rec}})$	genie
nominal	0.600	0.90	5.97×10^{-2}	5.97×10^{-2}	4.70×10^{-2}
C_1	0.440	0.95	1.30×10^{-2}	1.67×10^{-2}	6.81×10^{-3}
C_2	0.367	0.45	6.04×10^{-2}	2.68×10^{-2}	6.13×10^{-3}

S5.3 Ambiguity across links

Table S10 contains the data of Fig. 4 ($T = 60$, $\eta = 10^{-4}$) together with the worst channels, the mean burst length ℓ of the fit and the survival time D, the smallest at which the Gilbert fit certifies $R_{\mathrm{nom}} \le 10^{-4}$ with dense attempts.

TABLE S10
AMBIGUITY ACROSS LINKS

loss	ℓ	D	R_{nom}	R^+/R_{nom}	worst ρ	R_{nom}, B=15	R^+/R_{nom}, B=15	worst (ρ, b), B=15
0.5%	1.25	6	7.1×10^{-5}	68	0.816	2.8×10^{-3}	4.1	(0.625, 0.27)
0.5%	1.5	8	8.2×10^{-5}	69	0.875	7.8×10^{-4}	10	(0.325, 0.03)
0.5%	2	12	6.1×10^{-5}	86	0.925	2.2×10^{-4}	30	(0.900, 0.28)
0.5%	2.5	15	7.4×10^{-5}	67	0.950	1.7×10^{-4}	40	(0.950, 0.05)
0.5%	3	18	7.6×10^{-5}	61	0.962	1.3×10^{-4}	39	(0.937, 0.06)
0.5%	5	28	8.9×10^{-5}	46	0.994	1.1×10^{-4}	41	(0.981, 0.10)
1%	1.25	7	2.8×10^{-5}	285	0.850	1.2×10^{-3}	14	(0.750, 0.26)
1%	1.5	9	5.3×10^{-5}	182	0.891	5.7×10^{-4}	25	(0.725, 0.01)
1%	2	13	6.0×10^{-5}	162	0.933	2.1×10^{-4}	61	(0.900, 0.02)
1%	2.5	16	8.7×10^{-5}	107	0.955	2.0×10^{-4}	53	(0.933, 0.27)
1%	3	19	9.9×10^{-5}	89	0.967	1.8×10^{-4}	59	(0.950, 0.03)
1%	5	31	8.4×10^{-5}	95	1.000	1.0×10^{-4}	81	(0.987, 0.05)
2%	1.25	7	5.6×10^{-5}	285	0.850	3.0×10^{-3}	12	(0.750, 0.25)
2%	1.5	10	3.5×10^{-5}	493	0.908	4.3×10^{-4}	62	(0.850, 0.26)
2%	2	14	5.9×10^{-5}	305	0.942	2.3×10^{-4}	95	(0.925, 0.26)
2%	2.5	18	6.0×10^{-5}	278	0.961	1.5×10^{-4}	131	(0.950, 0.26)
2%	3	21	8.4×10^{-5}	193	0.975	1.6×10^{-4}	124	(0.950, 0.01)
2%	5	34	7.9×10^{-5}	203	1.000	9.9×10^{-5}	171	(0.987, 0.02)
5%	1.25	8	2.7×10^{-5}	$1.2k$	0.875	2.9×10^{-3}	22	(0.800, 0.00)
5%	1.5	10	8.7×10^{-5}	490	0.908	1.6×10^{-3}	42	(0.850, 0.25)
5%	2	15	7.2×10^{-5}	579	0.950	3.7×10^{-4}	156	(0.925, 0.00)
5%	2.5	19	8.8×10^{-5}	451	0.967	2.7×10^{-4}	185	(0.933, 0.00)
5%	3	23	8.9×10^{-5}	426	0.979	2.0×10^{-4}	249	(0.987, 0.01)
5%	5	37	9.1×10^{-5}	440	1.000	1.3×10^{-4}	341	(0.987, 0.01)
10%	1.25	8	5.5×10^{-5}	$1.2k$	0.875	8.7×10^{-3}	14	(0.825, 0.00)
10%	1.5	11	5.7×10^{-5}	$1.3k$	0.916	1.9×10^{-3}	69	(0.875, 0.00)
10%	2	16	7.0×10^{-5}	$1.1k$	0.958	5.7×10^{-4}	170	(0.000, 0.00)
10%	2.5	21	6.1×10^{-5}	$1.2k$	0.972	2.8×10^{-4}	370	(0.983, 0.00)
10%	3	25	7.5×10^{-5}	954	0.987	2.4×10^{-4}	372	(0.975, 0.00)
10%	5	40	8.3×10^{-5}	961	1.000	1.4×10^{-4}	574	(0.994, 0.00)
20%	1.25	9	2.1×10^{-5}	$5.3k$	0.891	1.1×10^{-2}	14	(0.825, 1.00)
20%	1.5	12	3.7×10^{-5}	$3.7k$	0.933	3.6×10^{-3}	48	(0.900, 0.25)
20%	2	17	6.9×10^{-5}	$2.1k$	0.958	1.3×10^{-3}	172	(0.950, 0.00)
20%	2.5	22	7.1×10^{-5}	$2.0k$	0.978	6.6×10^{-4}	240	(0.950, 0.00)
20%	3	26	9.8×10^{-5}	$1.4k$	0.987	5.5×10^{-4}	291	(0.975, 0.00)
20%	5	42	9.8×10^{-5}	$1.6k$	1.000	2.5×10^{-4}	638	(1.000, 0.00)

S5.4 Fading channels

Generation. Rayleigh fading with Clarke autocorrelation $J_0(2\pi f_D|t-t'|)$ between slots t and t' is generated by the inverse-DFT method of [1] on blocks of 2^{23} slots; the empirical autocorrelation matches J_0 to three decimals. A packet is lost if $|h|^2 < -\ln(1-\bar{e})$, which gives the loss rate $\bar{e}$. For composite fading, $|h|^2$ is multiplied by $10^{X_t/10}$, where X_t is a Gauss–Markov process with standard deviation 6 dB and correlation time 200 slots, and the threshold is the empirical $\bar{e}$-quantile of the received power. Four independent blocks are used per channel (3.4×10^7 slots).

Evaluation. Missions of $T = 60$ slots start every third slot; the trace-driven risk is the fraction of failing missions. Order-L fits use $(L{+}1)$-block frequencies of the same trace with a pseudo-count of 0.5. Table S11 gives all configurations. Its ratios are predicted/true risks for order-L Markov fits to the trace; $g{=}2$ denotes periodic attempts every second slot, and the last column gives the Gilbert and order-6 predictions for π_L ($B = 15$), which is designed on the Gilbert fit. Fig. S3 explains the errors through the run persistences of Proposition 1: the Gilbert fit extrapolates the first persistence (dotted lines), which overstates the risk where persistence first falls with depth (Rayleigh) and understates it where persistence grows with depth (shadowing and the measured links).

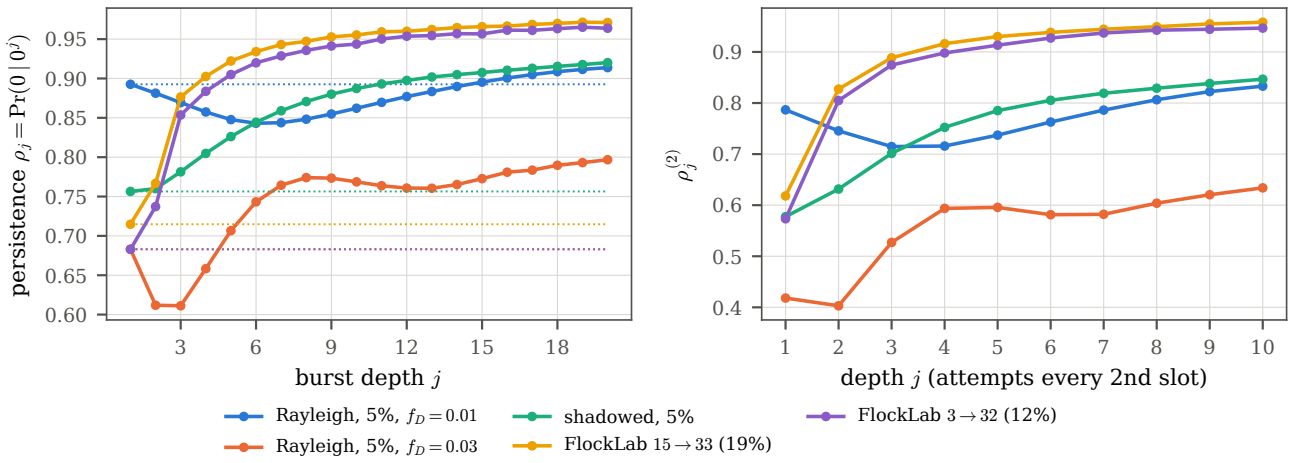


Fig. S3. Run persistences $\rho_j = \Pr(0 \mid 0^j)$ (left) and those of the 2-decimated channel (right) for fading channels and two measured links. Dotted lines: the constant persistence implied by the Gilbert fit.

TABLE S11
ALL FADING CONFIGURATIONS

channel	D	R dense	$L{=}1$	2	3	4	6	$M{=}2$ class	R, $g{=}2$	$L{=}1$	R, π_L	$L{=}1/L{=}6$
Rayleigh, 5%, $f_D = 0.01$	12	6.5×10^{-2}	1.3	1.3	1.2	1.2	1.2	$[4.5\times10^{-2}, 9.3\times10^{-2}]$	7.3×10^{-2}	1.3	8.4×10^{-2}	1.2 / 1.1
Rayleigh, 5%, $f_D = 0.01$	16	3.2×10^{-2}	1.6	1.5	1.4	1.2	1.1	$[3.8\times10^{-2}, 7.0\times10^{-2}]$	3.5×10^{-2}	1.6	3.7×10^{-2}	1.6 / 1.1
Rayleigh, 5%, $f_D = 0.01$	20	1.8×10^{-2}	1.7	1.5	1.3	1.1	0.93	$[2.4\times10^{-2}, 5.7\times10^{-2}]$	1.9×10^{-2}	1.7	2.0×10^{-2}	1.7 / 1
Rayleigh, 5%, $f_D = 0.03$	6	8.3×10^{-2}	1.5	1.2	1.2	1.2	1	$[3.4\times10^{-2}, 1.6\times10^{-1}]$	1.1×10^{-1}	1.5	2.1×10^{-1}	1.1 / 1
Rayleigh, 5%, $f_D = 0.03$	8	4.2×10^{-2}	1.4	0.89	0.89	1.1	1.1	$[1.8\times10^{-2}, 1.1\times10^{-1}]$	5.0×10^{-2}	1.5	7.5×10^{-2}	1.5 / 1
Rayleigh, 5%, $f_D = 0.03$	10	2.5×10^{-2}	1.1	0.54	0.54	0.74	0.99	$[5.5\times10^{-3}, 8.7\times10^{-2}]$	2.9×10^{-2}	1.2	3.7×10^{-2}	1.4 / 1.1
Rayleigh, 1%, $f_D = 0.005$	8	2.8×10^{-2}	1	1	1	1	1	$[8.7\times10^{-3}, 2.9\times10^{-2}]$	3.2×10^{-2}	0.96	3.9×10^{-2}	0.88 / 1
Rayleigh, 1%, $f_D = 0.005$	11	1.4×10^{-2}	1.4	1.3	1.2	1.2	1.1	$[8.7\times10^{-3}, 2.0\times10^{-2}]$	1.9×10^{-2}	1.2	1.8×10^{-2}	1.2 / 1.1
Rayleigh, 1%, $f_D = 0.005$	14	7.1×10^{-3}	1.7	1.5	1.3	1.2	1.1	$[8.3\times10^{-3}, 1.5\times10^{-2}]$	7.9×10^{-3}	1.6	9.0×10^{-3}	1.5 / 1.1
shadowed, 5%, $f_D = 0.03$	10	4.9×10^{-2}	1.1	1.1	1.2	1.3	1.3	$[2.5\times10^{-2}, 9.6\times10^{-2}]$	5.4×10^{-2}	1.2	6.4×10^{-2}	1.3 / 1.3
shadowed, 5%, $f_D = 0.03$	14	2.7×10^{-2}	0.6	0.62	0.78	0.94	1.2	$[5.5\times10^{-3}, 6.7\times10^{-2}]$	3.0×10^{-2}	0.66	3.3×10^{-2}	0.76 / 1.2
shadowed, 5%, $f_D = 0.03$	18	1.6×10^{-2}	0.3	0.32	0.45	0.62	0.92	$[1.1\times10^{-3}, 5.3\times10^{-2}]$	1.8×10^{-2}	0.34	1.9×10^{-2}	0.4 / 0.98
shadowed, 1%, $f_D = 0.03$	8	6.6×10^{-3}	0.44	0.71	1	1.2	1.2	$[4.2\times10^{-7}, 1.8\times10^{-2}]$	7.9×10^{-3}	0.56	1.1×10^{-2}	0.82 / 1.1
shadowed, 1%, $f_D = 0.03$	12	2.5×10^{-3}	0.086	0.21	0.46	0.73	1.1	$[5.4\times10^{-11}, 1.1\times10^{-2}]$	2.9×10^{-3}	0.11	3.5×10^{-3}	0.19 / 1.1
shadowed, 1%, $f_D = 0.03$	16	1.0×10^{-3}	0.015	0.053	0.18	0.39	0.81	$[7.0\times10^{-15}, 8.3\times10^{-3}]$	1.2×10^{-3}	0.02	1.4×10^{-3}	0.034 / 0.87

S5.5 Measured links

Data. We use the FlockLab link-quality dataset [2] (July–September 2019, TelosB, 802.15.4 channel 26, 0 dBm, 8 B payload). In each test every node sends 100 strobes, and every other node records a success/loss bit per strobe. The preprocessed files repeat each (test, sender, receiver) record several times (median 6.4×, always with identical content), so records are deduplicated on that key. Rows with a missing or truncated bit string and streams without any success (link down) are discarded. We keep the 171 directed links with loss rate in $[1\%, 25\%]$ and at least 500 distinct tests (in fact 567–805).

Evaluation. Each test is a short trace; missions of $T = 60$ slide over its 41 possible start positions. Fits and recovery curves are estimated from within-stream counts only. For each link, D is the smallest survival time at which the Gilbert fit certifies $R_{\text{nom}} \le 10^{-2}$ with dense attempts, and π_L ($B = 15$) is designed on that fit. Standard errors treat tests as independent units; the median relative standard error of the dense risk is 0.29, and the measured risk exceeds the Gilbert prediction by more than two standard errors on 47% of the links. Fig. S4 shows the whole population, and Table S12 (ℓ: mean burst length) the burstiest link of each loss band (1–4, 4–8, 8–15 and 15–25%). Its ratios are predicted/true; the last column gives the Gilbert-predicted ratio

for π_L ($B = 15$) and the measured risk of the recovery-curve redesign. On all four links, the measured dense risk lies within $[R^-, R^+]$ of the $M = 2$ class ($\eta = 10^{-4}$). The error grows with loss rate and burstiness; the second-order fit underestimates on more links than the Gilbert fit (85% versus 64%), and the order-6 fit overshoots by 1.5–2.1×. The persistence profile, pooled over tests, grows with burst depth (Fig. S3), as in the shadowed channels. Part of this growth may reflect differences between tests (slow regimes) rather than memory within a test; separating the two would require fits per test or per cluster of tests.

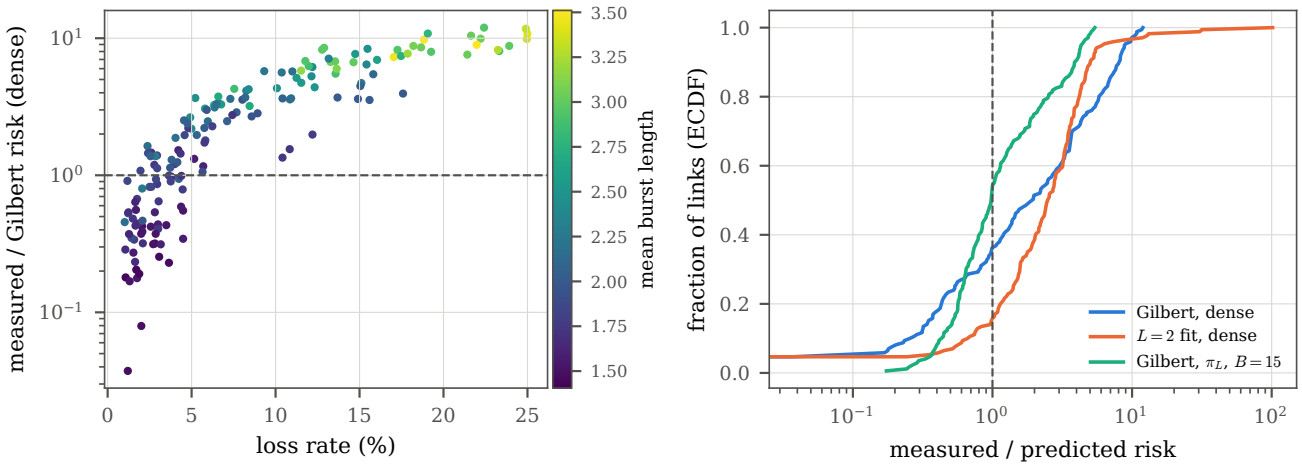


Fig. S4. Measured links (FlockLab, 171 TelosB links). Left: ratio of measured to Gilbert-predicted dense risk versus loss rate, colored by mean burst length. Right: empirical distribution of measured/predicted risk for the Gilbert fit (dense), the second-order fit (dense) and the Gilbert-predicted risk of π_L ($B = 15$).

TABLE S12
REPRESENTATIVE MEASURED LINKS

link	tests	loss	ℓ	D	R dense	L=1	2	3	4	6	R, π_L	L=1 / rec.
$15 \to 33$	718	18.8%	3.51	18	8.0×10^{-2}	0.1	0.26	1.1	1.3	1.5	1.1×10^{-1}	0.18 / 1.0×10^{-1}
$23 \to 22$	759	7.5%	2.86	13	3.2×10^{-2}	0.23	0.51	1.7	1.8	1.6	4.7×10^{-2}	0.37 / 4.5×10^{-2}
$27 \to 22$	801	2.9%	2.28	9	9.0×10^{-3}	0.73	1.2	3.5	3.2	2.1	1.5×10^{-2}	1.3 / 1.6×10^{-2}
$3 \to 32$	752	11.5%	3.16	15	5.0×10^{-2}	0.17	0.39	1.3	1.6	1.6	7.5×10^{-2}	0.26 / 7.1×10^{-2}

S6. REPRODUCIBILITY

All results are computed by exact recursions over the joint chain of channel context and controller state, in floating point or exact rational arithmetic, except the trace-driven risks. Table S13 maps the claims to scripts in `code/`. The script `verify_paper_claims.py` rechecks the key numerical claims of the main paper from the stored results, and `make_figs.py` followed by `make_supplement.py` regenerates this document's tables and data figures. Designing π_L for $T = 60$, $B = 15$ takes about 0.5 s in exact arithmetic, and evaluating one channel about 12 ms (Python, one core). The measured traces are downloaded by `get_flocklab.py` (about 50 MB).

TABLE S13
CLAIMS AND THE SCRIPTS THAT PRODUCE THEM

Claim	Script (`code/`)
Lemma 1, Eq. (12), Theorem 2, Fig. 2, Fig. S2, Tables S2 and S3	`exp_dense.py`, `exp_dense_M3.py`, `exp_phase.py`
Proposition 1, decimated persistences	`exp_decimation.py`
Eq. (14), Proposition S2, reversal criterion	`exp_analytic.py`
Eq. (15), Proposition S3, Table S1	`exp_closedloop.py`
Proposition 2, exact fractions	`exp_reversal.py`
Fig. 3, Table S4, b-sensitivity, bound tightness	`exp_budget.py`
Baselines, Table S5; η, Table S6; Remark S1	`exp_baselines.py`, `exp_eta.py`, `exp_M3L1.py`
Gilbert–Elliott channels, Table S8	`ge.py`, `exp_ge.py`
Redesign, Table S9	`exp_redesign.py`
Fig. 4, Table S10	`exp_sweep.py`
Table I (fading), Table S11, Fig. S3	`exp_fading.py` (argument `shadow`)
Table I (measured), Fig. S4, Table S12	`get_flocklab.py`, `exp_flocklab.py`
Table S7	`exp_mc_check.py`

REFERENCES

[1] D. J. Young and N. C. Beaulieu, "The generation of correlated Rayleigh random variates by inverse discrete Fourier transform," *IEEE Trans. Commun.*, vol. 48, no. 7, pp. 1114–1127, 2000.

[2] R. Jacob *et al.*, "Dataset: Wireless link quality estimation on FlockLab – and beyond," in *Proc. 2nd Workshop Data Acquisition To Analysis (DATA)*, 2019, pp. 57–60.